

Lecture 8: Asymptotic Distribution of OLS, Heteroskedasticity, and Robust Standard Errors

This next part of the course studies inference in OLS. Now that we know how to estimate linear models with OLS, understanding inference will allow us to construct confidence intervals for our estimates and test hypotheses about the coefficients. In order to do inference, we need to know what the distribution of our estimator and/or test statistic would be under a given null hypothesis.

Today, we will study the asymptotic distribution of OLS. In large samples, we can show that OLS is normally distributed, under the same assumptions that we used for consistency of OLS. We will also show how to estimate the standard error of $\hat{\beta}$. This will be the key to conducting hypothesis tests and constructing confidence intervals. We will also explore the ideas of homoskedasticity and heteroskedasticity, and how to construct standard errors that are robust to heteroskedasticity.

Note that we are focusing on the asymptotic distribution rather than the small-sample distribution. This is because we can show asymptotic normality of OLS without knowing the exact distribution of the error term, ε_i . If we were willing to assume that ε_i was normally distributed, then we could derive the finite-sample distribution of OLS. This finite-sample approach was important in the past when sample sizes were small. The problem with the finite-sample approach is that there is no reason to believe that the residual is normally distributed in real-world applications. Fortunately, modern data analysis typically has larger samples, and so we can confidently rely on asymptotic normality instead.

Asymptotic Normality of OLS

To prove that OLS is asymptotically normal, we will use the same assumptions that we used to show that OLS is consistent. These assumptions are:

Assumption (No Perfect Collinearity). *None of the regressors are perfectly collinear: it is not possible to perfectly predict one of the regressors as a linear combination of the other regressors and a constant.*

Assumption (IID Sampling). *Each observation is drawn independently and from the identical distribution as the other observations.*

Assumption (Orthogonality Condition). *The error term, ε_i , has zero mean $\mathbb{E}[\varepsilon_i] = 0$, and is uncorrelated with the regressors, $\text{Cov}(X_i^k, \varepsilon_i) = 0$ for each k .*

These assumptions will play a similar role to their role in the consistency proof. No perfect collinearity ensures that the OLS estimator has a unique solution. IID sampling allows us to use the law of large numbers and central limit theorem: we can show that $\hat{\beta}$ is converging to something, and that it is converging to a normal distribution around that number. The orthogonality condition ensures that the number $\hat{\beta}$ converges to is in fact the true β .

We will state the theorem in general, and then we will prove it for the single regressor case.

Theorem (OLS is Asymptotically Normal). *Suppose that the regressors are not perfectly collinear, each observation is sampled iid, and the orthogonality condition holds ($\mathbb{E}[\varepsilon_i] = 0$ and $\text{Cov}(X_{k,i}, \varepsilon_i) = 0$ for each k). Then, OLS is asymptotically normal, meaning that $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$. For the single regressor case, we have $V = \frac{\text{Var}((X_i - \mathbb{E}[X_i])\varepsilon_i)}{\text{Var}(X_i)^2}$.*

Before we begin, notice the way that we have written the result, subtracting off β and then multiplying by $\sqrt{N}$. If we had not multiplied by $\sqrt{N}$, we would just have $(\hat{\beta} - \beta) \xrightarrow{p} 0$, since we showed previously that $\hat{\beta}$ is consistent (it converges to the true β). We multiply specifically by $\sqrt{N}$ because OLS converges at rate $\frac{1}{\sqrt{N}}$: the variance is shrinking at rate $\frac{1}{N}$ (just like the sample mean), so multiplying by $\sqrt{N}$ will keep the variance constant. Given that we are multiplying by $\sqrt{N}$, we need to first subtract off β , otherwise we

would have $\sqrt{N} \cdot \hat{\beta}$, which diverges to infinity. You can interpret all of this as saying that $\hat{\beta}$ converges to a normal distribution, centered around the true β and with a standard error that shrinks at rate $\frac{1}{\sqrt{N}}$.

We will provide a sketch of the proof for the single regressor case. We are running the regression model:

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$

with the orthogonality conditions

$$\begin{aligned}\mathbb{E}[\varepsilon_i] &= 0 \\ \text{Cov}(X_i, \varepsilon_i) &= 0\end{aligned}$$

We have previously shown that

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(X_i, Y_i)}{\widehat{\text{Var}}(X_i)} = \beta + \frac{\widehat{\text{Cov}}(X_i, \varepsilon_i)}{\widehat{\text{Var}}(X_i)}$$

Note, importantly, that although $\text{Cov}(X_i, \varepsilon_i) = 0$ in the *population*, the term $\widehat{\text{Cov}}(X_i, \varepsilon_i)$ refers to the *sample covariance*. Thus, although $\widehat{\text{Cov}}(X_i, \varepsilon_i)$ will converge to zero as the sample size grows large, it will not quite equal zero in finite samples. This is what creates the variance in the sampling distribution of $\hat{\beta}$.

We can then subtract β and multiply by $\sqrt{N}$:

$$\sqrt{N}(\hat{\beta} - \beta) = \frac{\sqrt{N}\widehat{\text{Cov}}(X_i, \varepsilon_i)}{\widehat{\text{Var}}(X_i)}$$

We will not do the remainder of the proof in class, but we outline the steps below:

- We show that the denominator converges to the true variance, $\widehat{\text{Var}}(X_i) \xrightarrow{p} \text{Var}(X_i)$. This uses the assumption of iid sampling to apply the law of large numbers, plus Slutsky's Theorem.
- We show that the numerator converges to a normal, $\sqrt{N}\widehat{\text{Cov}}(X_i, \varepsilon_i) \xrightarrow{d} N(0, \text{Var}((X_i - \mathbb{E}[X_i])\varepsilon_i))$. This uses the orthogonality condition to show that the covariance is mean zero, and then uses iid sampling to apply a combination of the central limit theorem and the law of large numbers.
- We then use Slutsky's Theorem to show that the ratio converges in distribution to a normal distribution, $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$, where $V = \frac{\text{Var}((X_i - \mathbb{E}[X_i])\varepsilon_i)}{\text{Var}(X_i)^2}$.

This is a very useful result, but to apply it, we need to know the standard error of $\hat{\beta}$. We will next show how we can construct a consistent estimator of the standard error.

Homoskedasticity and Heteroskedasticity

Before we continue, we will introduce the concepts of homoskedasticity and heteroskedasticity.

Definition. We say that the error term, ε_i , is homoskedastic if $\mathbb{E}[\varepsilon_i^2 \mid X_{1,i}, \dots, X_{K,i}] = \mathbb{E}[\varepsilon_i^2]$.

Note that if the error term also has zero conditional mean ($\mathbb{E}[\varepsilon_i \mid X_{1,i}, \dots, X_{K,i}] = 0$), then this is equivalent to saying that the variance of ε_i does not depend on X_i . We will use the version defined in terms of $\mathbb{E}[\varepsilon_i^2 \mid X_{1,i}, \dots, X_{K,i}]$, since that is what we will need for our analysis.

Heteroskedasticity is just the opposite of homoskedasticity:

Definition. We say that the error term, ε_i , is heteroskedastic if it is not homoskedastic.

The exercises below ask you to identify a few examples where the error term may be heteroskedastic.

For each of the following exercises, consider a linear model $Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$, and explain whether the error term is likely to be homoskedastic or heteroskedastic.

Exercise 1. Suppose the outcome, Y_i , is an indicator, which can only be either zero or one. *Hint: If $Y_i = 1$ with probability p , then $\mathbb{E}[Y_i] = p$, and $\text{Var}(Y_i) = p(1-p)$.*

Exercise 2. Suppose that the unit of observation is a city, Y_i is the city-level unemployment rate, computed with a 1% sample of the city, and X_i is the city's population.

Exercise 3. Suppose that X_i is an indicator for receiving a drug in a randomized controlled trial, and the drug has heterogeneous effects: for some members of the population the drug is beneficial, and for others it is harmful.

It turns out that if the error term is homoskedastic, we can simplify our standard error dramatically. For notational simplicity, define $\tilde{X}_i = X_i - \mathbb{E}[X_i]$. We have:

$$\begin{aligned} \text{Var}((X_i - \mathbb{E}[X_i])\varepsilon_i) &= \text{Var}(\tilde{X}_i\varepsilon_i) \\ &= \mathbb{E}[(\tilde{X}_i\varepsilon_i - \mathbb{E}[\tilde{X}_i\varepsilon_i])^2] \\ &= \mathbb{E}[(\tilde{X}_i\varepsilon_i)^2] \\ &= \mathbb{E}[\mathbb{E}[(\tilde{X}_i\varepsilon_i)^2 | X_i]] \\ &= \mathbb{E}[\tilde{X}_i^2 \mathbb{E}[\varepsilon_i^2 | X_i]] \end{aligned}$$

where the third line uses the fact that $\mathbb{E}[\tilde{X}_i\varepsilon_i] = 0$ by orthogonality, and the fourth line uses the law of iterated expectations. If we have homoskedasticity, then we can simplify this:

$$\begin{aligned} \mathbb{E}[\tilde{X}_i^2 \mathbb{E}[\varepsilon_i^2 | X_i]] &= \mathbb{E}[\tilde{X}_i^2 \mathbb{E}[\varepsilon_i^2]] \\ &= \mathbb{E}[\tilde{X}_i^2] \mathbb{E}[\varepsilon_i^2] \\ &= \text{Var}(X_i) \text{Var}(\varepsilon_i) \end{aligned}$$

where the last line uses the fact that $\mathbb{E}[\varepsilon_i] = 0$. Thus, under homoskedasticity, we have:

$$V_{\text{Homoskedastic}} = \frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i)}$$

Note also that, in the general case, we have:

$$V = \frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i)} + \frac{\text{Cov}((X_i - \mathbb{E}[X_i])^2, \varepsilon_i^2)}{\text{Var}(X_i)^2}$$

When is the Standard Error Large or Small?

We can use these formulas to understand when our standard error will be large or small. The standard error of $\hat{\beta}$ is:

$$\text{SE}(\hat{\beta}) = \sqrt{\frac{V}{N}}$$

In the homoskedastic special case, we have:

$$\text{Under Homoskedasticity: } \text{SE}(\hat{\beta}) = \sqrt{\frac{V_{\text{Homoskedastic}}}{N}} = \sqrt{\frac{1}{N} \cdot \frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i)}}$$

This reveals three important contributors to the standard error of $\hat{\beta}$. The standard error will get smaller as the sample size N gets larger. The standard error will be larger if the variance of the residual, $\text{Var}(\varepsilon_i)$, is larger. And finally, the standard error will be smaller if the variance of the regressor, $\text{Var}(X_i)$, is larger.

The general, heteroskedastic case, adds an additional complication:

$$\text{SE}(\hat{\beta}) = \sqrt{\frac{V}{N}} = \sqrt{\frac{1}{N} \cdot \left(\frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i)} + \frac{\text{Cov}((X_i - \mathbb{E}[X_i])^2, \varepsilon_i^2)}{\text{Var}(X_i)^2} \right)}$$

The additional term, $\text{Cov}((X_i - \mathbb{E}[X_i])^2, \varepsilon_i^2) / \text{Var}(X_i)^2$ reflects the contribution of heteroskedasticity. In particular, observations where the regressor X_i is far from its mean are especially important for estimating the slope $\hat{\beta}$, while observations where X_i is near the mean are less important. If the error term has especially high variance when X_i is far from the mean, then heteroskedasticity will raise the standard error relative to the homoskedastic case. In contrast, if the error term has especially high variance when X_i is close to the mean, then heteroskedasticity will lower the standard error relative to the homoskedastic case.

Estimating the Variance of $\hat{\beta}$

We can now discuss how to consistently estimate V . If ε_i is homoskedastic, then a simple estimator would be $\frac{\widehat{\text{Var}}(\hat{\varepsilon}_i)}{\widehat{\text{Var}}(X_i)}$. That is:

$$\begin{aligned}\widehat{\text{Var}}(\hat{\varepsilon}_i) &= \frac{1}{N} \sum_{i=1}^N \hat{\varepsilon}_i^2 \\ \widehat{\text{Var}}(X_i) &= \overline{X^2} - \bar{X} \cdot \bar{X} \\ \hat{V}_{\text{Homoskedastic}} &= \frac{\widehat{\text{Var}}(\hat{\varepsilon}_i)}{\widehat{\text{Var}}(X_i)}\end{aligned}$$

Note that the sample mean of $\hat{\varepsilon}_i$ is zero by construction, thus we do not need to subtract the mean in our formula for $\widehat{\text{Var}}(\hat{\varepsilon}_i)$.¹

We have shown that the sample variance of X_i is a consistent estimator of the population variance many times now, so we will not repeat the proof here. For the sample variance of $\hat{\varepsilon}_i$, we know that the estimated residuals, $\hat{\varepsilon}_i = Y_i - \hat{\alpha} - \hat{\beta} \cdot X_i$, will converge to the true residuals because $\hat{\alpha} \xrightarrow{p} \alpha$, $\hat{\beta} \xrightarrow{p} \beta$. This, combined with the fact that the sample variance converges to the true population variance means that the sample variance of the estimated residuals will converge to the population variance of the true residuals, $\widehat{\text{Var}}(\hat{\varepsilon}_i) \xrightarrow{p} \text{Var}(\varepsilon_i)$.

Combining the fact that $\widehat{\text{Var}}(X_i) \xrightarrow{p} \text{Var}(X_i)$ and $\widehat{\text{Var}}(\hat{\varepsilon}_i) \xrightarrow{p} \text{Var}(\varepsilon_i)$, we have that $\hat{V}_{\text{Homoskedastic}} \xrightarrow{p} V_{\text{Homoskedastic}}$. So, if the error term is homoskedastic, then $\hat{V}_{\text{Homoskedastic}}$ will be a consistent estimator of V .

What if the error term is actually heteroskedastic? In that case, the homoskedastic standard error formula will typically not be consistent, since

$$V = \frac{\text{Var}((X_i - \mathbb{E}[X_i]) \varepsilon_i)}{\text{Var}(X_i)^2} = \frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i)} + \frac{\text{Cov}((X_i - \mathbb{E}[X_i])^2, \varepsilon_i^2)}{\text{Var}(X_i)^2}$$

¹It is common to include a "degrees-of-freedom" correction: if we multiply $\widehat{\text{Var}}(X_i)$ by $\frac{N}{N-1}$, then it will not just be consistent, it will also be an unbiased estimator. For $\widehat{\text{Var}}(\hat{\varepsilon}_i)$, we would multiply by $\frac{N}{N-2}$. In each case, the correction adjusts for the fact that a parameter has already been estimated, leading the sample variance to be slightly too low: for the sample variance of X_i , the mean of X_i needed to be estimated first (thus $N-1$), while for the sample variance of $\hat{\varepsilon}_i$, the coefficients $\hat{\alpha}$ and $\hat{\beta}$ had to be estimated first (thus $N-2$; if there were K regressors then our correction would be $N-K-1$). We do not need to do the degrees-of-freedom correction for consistency.

which will in general not equal $\frac{\text{Var}(\varepsilon_i)}{\text{Var}(X_i)}$. However, in the heteroskedastic case, we can again use sample analogues to the population moments. In particular:

$$\begin{aligned}\widehat{\text{Var}}((X_i - \bar{X})\hat{\varepsilon}_i) &= \frac{1}{N} \sum_{i=1}^N ((X_i - \bar{X})\hat{\varepsilon}_i)^2 \\ \widehat{\text{Var}}(X_i) &= \overline{X^2} - \bar{X} \cdot \bar{X} \\ \hat{V}_{\text{Robust}} &= \frac{\widehat{\text{Var}}((X_i - \bar{X})\hat{\varepsilon}_i)}{\widehat{\text{Var}}(X_i)^2}\end{aligned}$$

Note that $(X_i - \bar{X})\hat{\varepsilon}_i$ has a sample mean of zero by construction (this is the orthogonality condition that solves the OLS problem), so we do not need to subtract off its sample mean in the definition of the sample variance.

It is similarly possible to show that $\hat{V}_{\text{Robust}} \xrightarrow{p} V$. Note that this will be true whether the error is homoskedastic or heteroskedastic. This is why we refer to these as “robust standard errors.”

Which Standard Error Should We Use?

In principle, if we knew for sure that the error term was homoskedastic, then the homoskedastic standard errors are more efficient, meaning that they will generally give a more precise estimate of V . However, robust standard errors are still consistent, so they will still converge to the true standard error in a large sample.

In practice, there is almost never a good reason to believe that the error term is homoskedastic. Moreover, even if the error term is homoskedastic, the robust standard error will still give valid standard errors. **Thus, there is a broad consensus that you should always use robust standard errors.**

In Stata, the default is to give non-robust standard errors, and you need to specify `, rob` to get robust standard errors. Remember to always add `, rob` to the end of your regressions.

Exercise 4. Suppose that the error term is homoskedastic, and we have a large sample. Will the homoskedastic standard error, $\sqrt{\hat{V}_{\text{Homoskedastic}}/N}$, generally be larger, smaller, or around the same size as the robust standard error, $\sqrt{\hat{V}_{\text{Robust}}/N}$?

Bonus Material: Standard Errors with Multiple Regressors

How does this work for multiple regressors? We previously discussed the Frisch-Waugh-Lovell Theorem. To recap, that theorem told us that if we have the regression:

$$Y_i = \alpha + \beta_1 X_{1,i} + \dots + \beta_K X_{K,i} + \varepsilon_i$$

then we can obtain the OLS estimate of $\hat{\beta}_1$ by first “residualizing” X_1 and Y on the other regressors.² In particular, we define $\tilde{X}_{1,i}$ as the residual from a regression of $X_{1,i}$ on $X_{2,i}, \dots, X_{K,i}$, and $\tilde{Y}_i$ as the residual from a regression of Y_i on $X_{2,i}, \dots, X_{K,i}$. The Frisch-Waugh-Lovell Theorem tells us that

$$\hat{\beta}_1 = \frac{\widehat{\text{Cov}}(\tilde{X}_{1,i}, \tilde{Y}_i)}{\widehat{\text{Var}}(\tilde{X}_{1,i})}.$$

It turns out that in a multiple regression, we can use a similar trick to get the standard errors. In particular, we have:

$$\widehat{\text{SE}}_{\text{Robust}}(\hat{\beta}_1) = \sqrt{\frac{1}{N} \cdot \frac{\widehat{\text{Var}}(\tilde{X}_{1,i} \cdot \tilde{\varepsilon}_i)}{\widehat{\text{Var}}(\tilde{X}_{1,i})^2}}$$

²The FWL Theorem only requires you to residualize the regressor to get the correct point estimate: you can either residualize the outcome or not. For the standard errors, however, we will need to residualize both.

where we note that $\tilde{X}_{1,i}$ has zero sample mean by construction. Note also that $\hat{\varepsilon}_i$ will be the same whether you use $\tilde{Y}_i$ or Y_i as the outcome.³

This formula is also useful for thinking about how the standard error will change if we add in controls. For simplicity, we will assume homoskedasticity for the remainder of this discussion, so that $V = \frac{\text{Var}(\varepsilon_i)}{\text{Var}(\tilde{X}_{1,i})}$. Suppose that we have a regression of Y_i on $X_{1,i}$, and we are thinking of adding in a control, $X_{2,i}$. Suppose that we are in a situation where we know that the omitted variable bias is zero, we are just thinking of adding the control because it might improve our standard error. If the omitted variable bias is zero, then this means that either $\beta_2 = 0$ or $\text{Cov}(X_{1,i}, X_{2,i}) = 0$. In the first case, in a large sample, including the control will only increase the size of our standard error: ε_i will be unchanged, but the variance of $\tilde{X}_{1,i}$ will be smaller than the variance of $X_{1,i}$, so the standard error will get larger. Intuitively, the regression has trouble distinguishing between the effect of $X_{1,i}$ and $X_{2,i}$ when they are correlated. In contrast, when $\text{Cov}(X_{1,i}, X_{2,i}) = 0$, the variance of the residual will shrink, but $\text{Var}(\tilde{X}_{1,i}) = \text{Var}(X_{1,i})$. Thus, the standard error will get smaller: this is because the added regressor explains some of the error term, and thus reduces the remaining variance in the regression.

³If you generate $\tilde{Y}_i$ and $\tilde{X}_{1,i}$ and then run the regression with those residualized variables, Stata will give you the correct coefficient and also the correct standard error. There may be a small difference in the standard error compared to running the multiple regression, since Stata does a different degree of freedom correction depending on the number of regressors.